



SYMBIOSIS INSTITUTE OF TECHNOLOGY, NAGPUR

Symbiosis International (Deemed University)

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Practical No.: 9

Aim –

Implement localStorage and sessionStorage to save user preferences like theme settings.

Case Study: Create a Schedule Planner for the given table and show an alert box on click event.

Software/Tools Required –

- VSCode
- Chrome
- HTML5
- CSS3
- Javascript (ES6)

Experiment Program Code –

index.html

```
<!DOCTYPE html>
<html lang="en">
<head>
  <meta charset="UTF-8" />
  <meta name="viewport" content="width=device-width, initial-scale=1.0" />
  <title>Theme Preferences</title>

  <style>
    * {
      box-sizing: border-box;
    }

    body {
      margin: 0;
      min-height: 100vh;
      display: flex;
```

```
align-items: center;
justify-content: center;
font-family: Arial, sans-serif;
```

```
background:
  radial-gradient(circle at 15% 20%, #c7d2fe 0%, transparent 30%),
  radial-gradient(circle at 85% 80%, #fbcfe8 0%, transparent 30%),
  linear-gradient(135deg, #eef2ff, #f8fafc);
```

```
color: #1e293b;
transition: all 0.5s ease;
overflow: hidden;
}
```

```
body::before,
body::after {
  content: "";
  position: fixed;
  border-radius: 50%;
  filter: blur(2px);
  z-index: -1;
}
```

```
body::before {
  width: 220px;
  height: 220px;
  background: rgba(129, 140, 248, 0.25);
  top: -70px;
  left: -70px;
}
```

```
body::after {
  width: 260px;
  height: 260px;
  background: rgba(244, 114, 182, 0.2);
  bottom: -100px;
  right: -80px;
}
```

```
.container {
  width: 90%;
  max-width: 570px;
  padding: 50px 45px;
```

```
text-align: center;
```

```
background: rgba(255, 255, 255, 0.72);
backdrop-filter: blur(18px);
-webkit-backdrop-filter: blur(18px);
```

```
border: 1px solid rgba(255, 255, 255, 0.7);
border-radius: 28px;
```

```
box-shadow:
  0 25px 60px rgba(30, 41, 59, 0.15),
  inset 0 1px 0 rgba(255, 255, 255, 0.8);
```

```
transition: 0.5s ease;
}
```

```
.icon {
  width: 82px;
  height: 82px;
```

```
margin: 0 auto 22px;
```

```
display: flex;
```

```
align-items: center;
justify-content: center;

font-size: 38px;

border-radius: 24px;

background: linear-gradient(
  135deg,
  #6366f1,
  #8b5cf6,
  #ec4899
);

box-shadow:
  0 12px 30px rgba(99, 102, 241, 0.35);

animation: float 3s ease-in-out infinite;
}

@keyframes float {
  0%, 100% {
    transform: translateY(0);
  }

  50% {
    transform: translateY(-6px);
  }
}

h1 {
  margin: 0;

  font-size: 32px;
  font-weight: 700;

  background: linear-gradient(
    90deg,
    #4f46e5,
    #7c3aed,
    #db2777
  );

  -webkit-background-clip: text;
  -webkit-text-fill-color: transparent;
}

.subtitle {
  margin: 14px auto 34px;

  max-width: 430px;

  color: #64748b;
  font-size: 15px;
  line-height: 1.7;
}

.options {
  display: flex;
  flex-direction: column;
  gap: 14px;
}

button {
  width: 100%;
  padding: 15px 20px;
```

```
border: none;
border-radius: 14px;

font-size: 15px;
font-weight: 600;

cursor: pointer;

display: flex;
align-items: center;
justify-content: center;
gap: 10px;

transition: all 0.25s ease;
}

button:hover {
  transform: translateY(-3px);
}

button:active {
  transform: scale(0.98);
}

.light-btn {
  color: #3730a3;

  background: linear-gradient(
    135deg,
    #eef2ff,
    #e0e7ff
  );

  border: 1px solid #c7d2fe;
}

.light-btn:hover {
  box-shadow: 0 10px 25px rgba(99, 102, 241, 0.18);
}

.dark-btn {
  color: white;

  background: linear-gradient(
    135deg,
    #1e293b,
    #334155
  );

  box-shadow: 0 8px 20px rgba(15, 23, 42, 0.18);
}

.dark-btn:hover {
  box-shadow: 0 12px 28px rgba(15, 23, 42, 0.28);
}

.clear-btn {
  color: #be123c;

  background: rgba(255, 228, 230, 0.7);

  border: 1px solid #fecdd3;
}

.clear-btn:hover {
  background: #ffe4e6;
  box-shadow: 0 8px 20px rgba(244, 63, 94, 0.15);
}
```

```
}

.storage-info {
  margin-top: 28px;

  font-size: 12px;
  color: #94a3b8;
}

.storage-info span {
  font-weight: bold;
  color: #6366f1;
}

body.dark {
  background:
    radial-gradient(circle at 15% 20%, #312e81 0%, transparent 30%),
    radial-gradient(circle at 85% 80%, #701a75 0%, transparent 30%),
    linear-gradient(135deg, #020617, #0f172a);

  color: white;
}

body.dark .container {
  background: rgba(15, 23, 42, 0.72);

  border: 1px solid rgba(148, 163, 184, 0.15);

  box-shadow:
    0 25px 60px rgba(0, 0, 0, 0.45),
    inset 0 1px 0 rgba(255, 255, 255, 0.05);
}

body.dark h1 {
  background: linear-gradient(
    90deg,
    #818cf8,
    #a78bfa,
    #f472b6
  );

  -webkit-background-clip: text;
  -webkit-text-fill-color: transparent;
}

body.dark .subtitle {
  color: #cbd5e1;
}

body.dark .light-btn {
  color: #e0e7ff;

  background: rgba(79, 70, 229, 0.15);

  border: 1px solid rgba(129, 140, 248, 0.3);
}

body.dark .dark-btn {
  background: linear-gradient(
    135deg,
    #f8fafc,
    #e2e8f0
  );

  color: #0f172a;
}
```

```

body.dark .clear-btn {
  color: #fda4af;

  background: rgba(159, 18, 57, 0.18);

  border: 1px solid rgba(244, 63, 94, 0.25);
}

body.dark .storage-info {
  color: #64748b;
}

body.dark .storage-info span {
  color: #818cf8;
}

@media (max-width: 500px) {

  .container {
    padding: 38px 25px;
  }

  h1 {
    font-size: 27px;
  }

  .icon {
    width: 70px;
    height: 70px;
    font-size: 32px;
  }

  .subtitle {
    font-size: 14px;
  }
}
</style>
</head>

<body>

<div class="container">

  <div class="icon">
    🌈
  </div>

  <h1>Theme Preferences</h1>

  <p class="subtitle">
    Personalize your experience by choosing a theme.
    Your preference will be remembered automatically.
  </p>

  <div class="options">

    <button class="light-btn" onclick="setTheme('light')">
      ☀️
      <span>Light Theme</span>
    </button>

    <button class="dark-btn" onclick="setTheme('dark')">
      🌙
      <span>Dark Theme</span>
    </button>
  </div>
</div>

```

```
</button>

<button class="clear-btn" onclick="clearPreference()">
  ⌵
  <span>Clear Preference</span>
</button>

</div>

<div class="storage-info">
  Preference saved using
  <span>localStorage</span> &
  <span>sessionStorage</span>
</div>

</div>

<script>
function setTheme(theme) {
  if (theme === 'dark') {
    document.body.classList.add('dark');
  } else {
    document.body.classList.remove('dark');
  }

  localStorage.setItem('theme', theme);

  sessionStorage.setItem('sessionTheme', theme);
}

function clearPreference() {
  localStorage.removeItem('theme');
  sessionStorage.removeItem('sessionTheme');

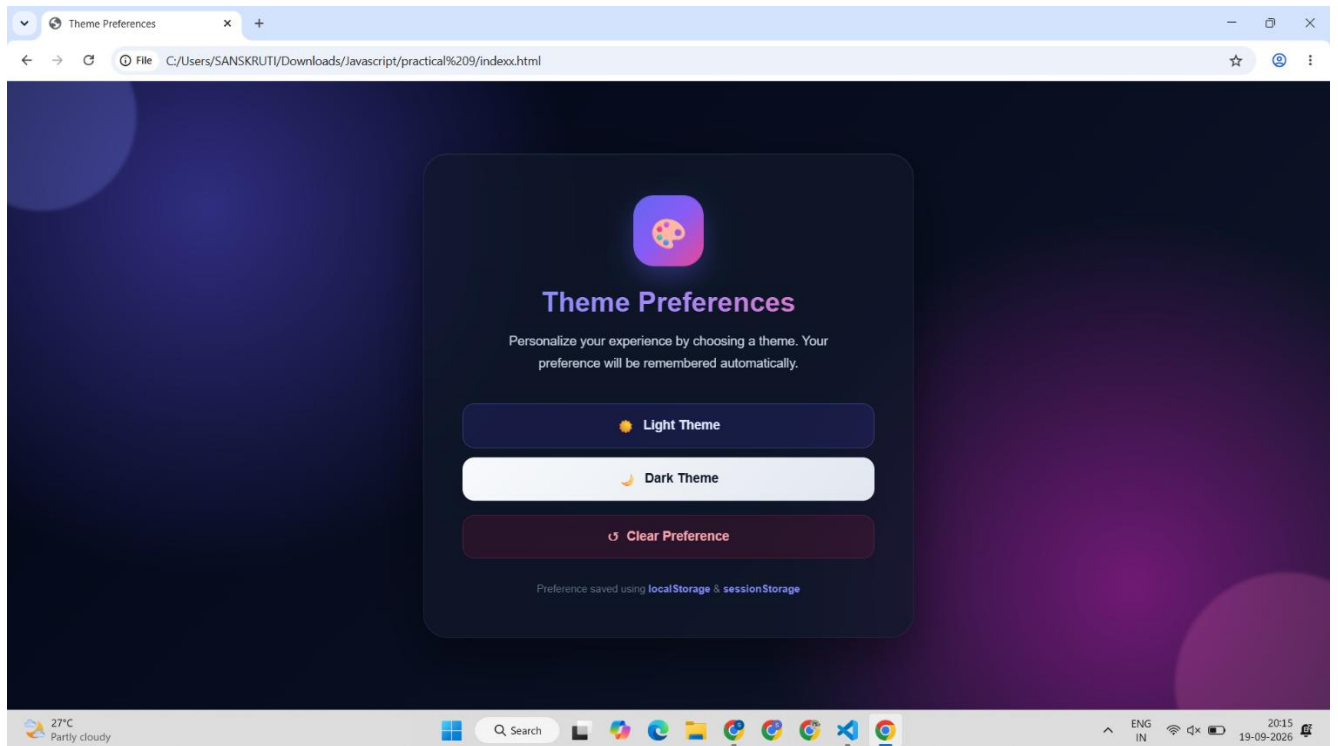
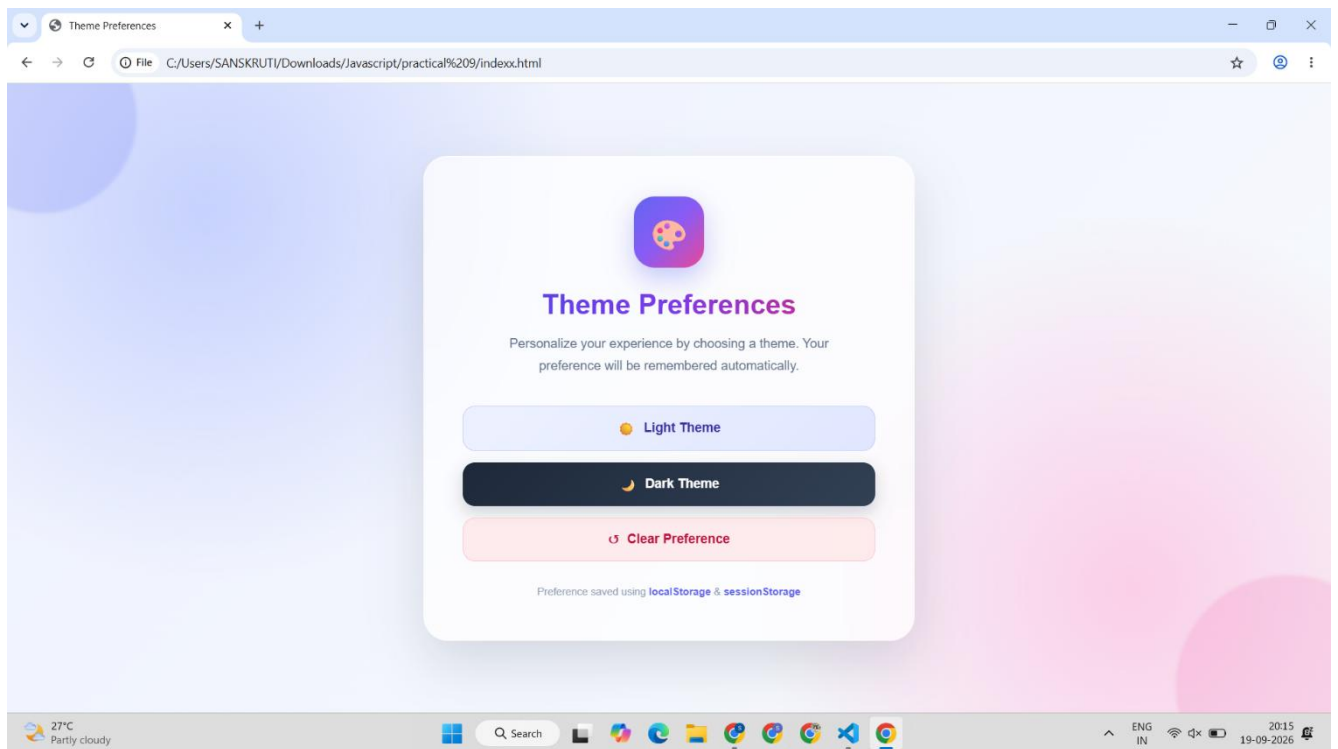
  document.body.classList.remove('dark');
}

window.onload = function () {
  var saved = localStorage.getItem('theme');

  if (saved === 'dark') {
    document.body.classList.add('dark');
  }
};
</script>

</body>
</html>
```

Output –



Case Study Title –

Create a Schedule Planner for the given table and show an alert box on click event.

Case Study Program Code –

index.html

```
<!DOCTYPE html>
<html>
<head>
  <title>Schedule Planner</title>

<style>
  * {
    box-sizing: border-box;
  }

  body {
    margin: 0;
    font-family: "Segoe UI", Arial, sans-serif;
    background: #f3f1ed;
    color: #252333;
    padding: 35px 20px;
  }

  .container {
    max-width: 1100px;
    margin: auto;
  }

  .top {
    display: grid;
    grid-template-columns: 1.05fr 2fr;
    gap: 18px;
    margin-bottom: 18px;
  }

  .brand {
    background: #29263d;
    border-radius: 24px;
    padding: 30px;
    color: white;
    min-height: 190px;
    display: flex;
    flex-direction: column;
    justify-content: space-between;
    position: relative;
    overflow: hidden;
  }

  .brand::before {
    content: "";
    position: absolute;
    width: 150px;
    height: 150px;
    border: 1px solid #ffffff20;
    border-radius: 50%;
    right: -50px;
    bottom: -50px;
  }

```

```
.brand::after {
  content: "";
  position: absolute;
  width: 80px;
  height: 80px;
  border: 1px solid #ffffff18;
  border-radius: 50%;
  right: 5px;
  bottom: 5px;
}

.brand small {
  color: #b9b3dc;
  text-transform: uppercase;
  letter-spacing: 2px;
  font-size: 10px;
  font-weight: 700;
}

.brand h1 {
  font-size: 32px;
  line-height: 1.05;
  margin: 8px 0;
  position: relative;
  z-index: 1;
}

.brand p {
  color: #c9c6d5;
  margin: 0;
  font-size: 13px;
  position: relative;
  z-index: 1;
}

.add-panel {
  background: #ffffff;
  border-radius: 24px;
  padding: 27px 30px;
  border: 1px solid #e5e1da;
}

.add-heading {
  display: flex;
  justify-content: space-between;
  align-items: center;
  margin-bottom: 20px;
}

.add-heading h2 {
  margin: 0;
  font-size: 18px;
}

.add-heading span {
  color: #928d9f;
  font-size: 11px;
}

.form-grid {
  display: grid;
  grid-template-columns: 1fr 1fr 1fr 1.4fr;
  gap: 11px;
}

.field label {
```

```
display: block;
color: #817d89;
font-size: 10px;
font-weight: 800;
text-transform: uppercase;
letter-spacing: .8px;
margin-bottom: 6px;
}

input {
width: 100%;
height: 43px;
border: 1px solid #dedbe3;
background: #faf9fb;
border-radius: 10px;
padding: 0 12px;
font-size: 13px;
color: #302d3c;
outline: none;
}

input:focus {
background: white;
border-color: #7668d8;
box-shadow: 0 0 0 3px #7668d815;
}

.add-btn {
height: 42px;
margin-top: 15px;
padding: 0 21px;
border: 0;
border-radius: 10px;
background: #7668d8;
color: white;
font-weight: 700;
font-size: 12px;
cursor: pointer;
}

.add-btn:hover {
background: #6557c5;
}

.schedule {
background: white;
border-radius: 24px;
padding: 28px;
border: 1px solid #e5e1da;
}

.schedule-top {
display: flex;
justify-content: space-between;
align-items: end;
margin-bottom: 20px;
}

.schedule-top h2 {
margin: 0;
font-size: 21px;
}

.schedule-top p {
margin: 5px 0 0;
font-size: 12px;
color: #918d99;
```

```

}

.storage-badge {
  display: flex;
  gap: 7px;
}

.badge {
  padding: 6px 9px;
  border-radius: 7px;
  font-size: 9px;
  font-weight: 700;
}

.local {
  background: #eeeeff;
  color: #6255b9;
}

.session {
  background: #e8f5ef;
  color: #31805c;
}

.table-wrap {
  overflow-x: auto;
}

table {
  width: 100%;
  border-collapse: collapse;
}

th {
  text-align: left;
  padding: 11px 14px;
  color: #9995a0;
  font-size: 9px;
  text-transform: uppercase;
  letter-spacing: 1px;
  border-bottom: 1px solid #ece9ee;
}

th:last-child {
  text-align: center;
}

td {
  padding: 15px 14px;
  border-bottom: 1px solid #f0edf1;
  font-size: 13px;
  cursor: pointer;
  transition: .15s;
}

tbody tr:hover td {
  background: #fbfaff;
}

tbody tr:last-child td {
  border-bottom: 0;
}

.day {
  font-weight: 750;
  color: #302d3d;
}

```

```
.day span {
  display: inline-flex;
  align-items: center;
  gap: 8px;
}

.dot {
  width: 7px;
  height: 7px;
  background: #7668d8;
  border-radius: 50%;
}

.time {
  color: #656170;
  font-weight: 600;
}

.topic {
  color: #6558c4;
  font-weight: 700;
}

.missing {
  color: #cf6872;
  font-style: italic;
}

.actions {
  text-align: center;
  cursor: default;
}

.action {
  width: 32px;
  height: 32px;
  border: 0;
  border-radius: 9px;
  cursor: pointer;
  font-size: 13px;
  margin: 0 2px;
}

.edit {
  background: #eeeeff;
  color: #6255b9;
}

.delete {
  background: #fff0f1;
  color: #c75b66;
}

.edit:hover {
  background: #dfdaff;
}

.delete:hover {
  background: #ffe0e3;
}

.empty {
  text-align: center;
  padding: 40px;
  color: #aaa6b1;
  font-size: 13px;
}
```

```

}

.edit-box {
  display: none;
  margin-top: 20px;
  background: #f7f5ff;
  border: 1px solid #e2def9;
  border-radius: 15px;
  padding: 20px;
}

.edit-box h3 {
  margin: 0 0 16px;
  font-size: 15px;
  color: #50479b;
}

.edit-grid {
  display: grid;
  grid-template-columns: 1fr 1fr 1fr 1.4fr;
  gap: 11px;
}

.save,
.cancel {
  margin-top: 14px;
  height: 38px;
  border: 0;
  border-radius: 9px;
  padding: 0 17px;
  font-size: 11px;
  font-weight: 700;
  cursor: pointer;
}

.save {
  background: #3d9568;
  color: white;
}

.cancel {
  background: #e4e1e9;
  color: #55515e;
  margin-left: 5px;
}

@media (max-width: 800px) {
  .top {
    grid-template-columns: 1fr;
  }

  .form-grid,
  .edit-grid {
    grid-template-columns: 1fr 1fr;
  }
}

@media (max-width: 500px) {
  body {
    padding: 20px 12px;
  }

  .form-grid,
  .edit-grid {
    grid-template-columns: 1fr;
  }
}

```

```

        .brand {
            min-height: 160px;
        }

        .schedule {
            padding: 18px;
        }

        table {
            min-width: 700px;
        }
    }
</style>
</head>

<body>

<div class="container">

    <div class="top">

        <div class="brand">
            <div>
                <small>Seminar · 2026</small>
                <h1>Schedule<br>Planner</h1>
            </div>

            <p>Build your timetable one session at a time.</p>
        </div>

        <div class="add-panel">

            <div class="add-heading">
                <h2>Add a session</h2>
                <span>Enter all four details</span>
            </div>

            <div class="form-grid">

                <div class="field">
                    <label>Day</label>
                    <input id="day" placeholder="Monday">
                </div>

                <div class="field">
                    <label>Begin</label>
                    <input id="begin" placeholder="8:00 a.m.">
                </div>

                <div class="field">
                    <label>End</label>
                    <input id="end" placeholder="5:00 p.m.">
                </div>

                <div class="field">
                    <label>Topic</label>
                    <input id="topic" placeholder="Introduction to XML">
                </div>

            </div>

            <button class="add-btn" onclick="addSchedule()">
                + Add session
            </button>

        </div>
    </div>

```

</div>

<div class="schedule">

<div class="schedule-top">

<div>

<h2>Your timetable</h2>

<p>Click a day, time or topic to view session details.</p>

</div>

<div class="storage-badge">

LOCAL STORAGE

SESSION STORAGE

</div>

</div>

<div class="table-wrap">

<table>

<thead>

<tr>

<th>Day</th>

<th>Begin</th>

<th>End</th>

<th>Topic</th>

<th>Actions</th>

</tr>

</thead>

<tbody id="scheduleBody"></tbody>

</table>

</div>

<div class="edit-box" id="editBox">

<h3>◆ Edit session</h3>

<div class="edit-grid">

<div class="field">

<label>Day</label>

<input id="editDay">

</div>

<div class="field">

<label>Begin</label>

<input id="editBegin">

</div>

<div class="field">

<label>End</label>

<input id="editEnd">

</div>

<div class="field">

<label>Topic</label>

<input id="editTopic">

</div>

</div>

<button class="save" onclick="saveEdit()">

```

        Save changes
    </button>

    <button class="cancel" onclick="cancelEdit()">
        Cancel
    </button>

</div>

</div>

</div>

<script>

let editIndex = -1;

window.onload = function() {
    displaySchedules();
};

function addSchedule() {

    let day = document.getElementById("day").value.trim();
    let begin = document.getElementById("begin").value.trim();
    let end = document.getElementById("end").value.trim();
    let topic = document.getElementById("topic").value.trim();

    if (day === "" || begin === "" || end === "" || topic === "") {
        alert("Please enter all details.");
        return;
    }

    let schedules =
        JSON.parse(localStorage.getItem("scheduleData")) || [];

    schedules.push({
        day: day,
        begin: begin,
        end: end
    });

    localStorage.setItem(
        "scheduleData",
        JSON.stringify(schedules)
    );

    let topics =
        JSON.parse(sessionStorage.getItem("topicData")) || [];

    topics.push(topic);

    sessionStorage.setItem(
        "topicData",
        JSON.stringify(topics)
    );

    document.getElementById("day").value = "";
    document.getElementById("begin").value = "";
    document.getElementById("end").value = "";
    document.getElementById("topic").value = "";

    displaySchedules();

    alert("Session added successfully!");
}

```

```

function displaySchedules() {

    let schedules =
        JSON.parse(localStorage.getItem("scheduleData")) || [];

    let topics =
        JSON.parse(sessionStorage.getItem("topicData")) || [];

    let body =
        document.getElementById("scheduleBody");

    body.innerHTML = "";

    if (schedules.length === 0) {

        body.innerHTML = `
            <tr>
                <td colspan="5" class="empty">
                    No sessions yet. Add your first session above.
                </td>
            </tr>
        `;

        return;
    }

    schedules.forEach(function(schedule, index) {

        let row = document.createElement("tr");

        let day = document.createElement("td");
        day.className = "day clickable";
        day.innerHTML =
            `<span><i class="dot"></i>${schedule.day}</span>`;

        let begin = document.createElement("td");
        begin.className = "time";
        begin.innerText = schedule.begin;

        let end = document.createElement("td");
        end.className = "time";
        end.innerText = schedule.end;

        let topic = document.createElement("td");

        if (topics[index]) {
            topic.className = "topic";
            topic.innerText = topics[index];
        } else {
            topic.className = "missing";
            topic.innerText = "Topic unavailable";
        }

        let actions = document.createElement("td");
        actions.className = "actions";

        let edit = document.createElement("button");
        edit.className = "action edit";
        edit.innerText = "✎";
        edit.title = "Edit";

        edit.onclick = function(event) {
            event.stopPropagation();
            editSchedule(index);
        };

        let del = document.createElement("button");
    }

```

```

    del.className = "action delete";
    del.innerText = "×";
    del.title = "Delete";

    del.onclick = function(event) {
        event.stopPropagation();
        deleteSchedule(index);
    };

    actions.appendChild(edit);
    actions.appendChild(del);

    row.appendChild(day);
    row.appendChild(begin);
    row.appendChild(end);
    row.appendChild(topic);
    row.appendChild(actions);

    [day, begin, end, topic].forEach(function(cell) {

        cell.onclick = function() {

            alert(
                "SEMINAR DETAILS\n\n" +
                "Day: " + schedule.day + "\n" +
                "Begin: " + schedule.begin + "\n" +
                "End: " + schedule.end + "\n" +
                "Topic: " + (topics[index] || "Topic unavailable")
            );

        };

    });

    body.appendChild(row);
});
}

function editSchedule(index) {

    let schedules =
        JSON.parse(localStorage.getItem("scheduleData")) || [];

    let topics =
        JSON.parse(sessionStorage.getItem("topicData")) || [];

    editIndex = index;

    document.getElementById("editDay").value =
        schedules[index].day;

    document.getElementById("editBegin").value =
        schedules[index].begin;

    document.getElementById("editEnd").value =
        schedules[index].end;

    document.getElementById("editTopic").value =
        topics[index] || "";

    document.getElementById("editBox").style.display =
        "block";
}

function saveEdit() {

    if (editIndex === -1) {

```

```

    return;
}

let day =
    document.getElementById("editDay").value.trim();

let begin =
    document.getElementById("editBegin").value.trim();

let end =
    document.getElementById("editEnd").value.trim();

let topic =
    document.getElementById("editTopic").value.trim();

if (day === "" || begin === "" || end === "" || topic === "") {
    alert("Please enter all details.");
    return;
}

let schedules =
    JSON.parse(localStorage.getItem("scheduleData")) || [];

schedules[editIndex] = {
    day: day,
    begin: begin,
    end: end
};

localStorage.setItem(
    "scheduleData",
    JSON.stringify(schedules)
);

let topics =
    JSON.parse(sessionStorage.getItem("topicData")) || [];

topics[editIndex] = topic;

sessionStorage.setItem(
    "topicData",
    JSON.stringify(topics)
);

editIndex = -1;

document.getElementById("editBox").style.display =
    "none";

displaySchedules();

alert("Session updated successfully!");
}

function cancelEdit() {
    editIndex = -1;

    document.getElementById("editBox").style.display =
        "none";
}

function deleteSchedule(index) {
    if (!confirm("Delete this session?")) {
        return;
    }
}

```

```

let schedules =
  JSON.parse(localStorage.getItem("scheduleData")) || [];

schedules.splice(index, 1);

localStorage.setItem(
  "scheduleData",
  JSON.stringify(schedules)
);

let topics =
  JSON.parse(sessionStorage.getItem("topicData")) || [];

topics.splice(index, 1);

sessionStorage.setItem(
  "topicData",
  JSON.stringify(topics)
);

displaySchedules();

alert("Session deleted successfully!");
}

</script>

</body>
</html>

```

Output –

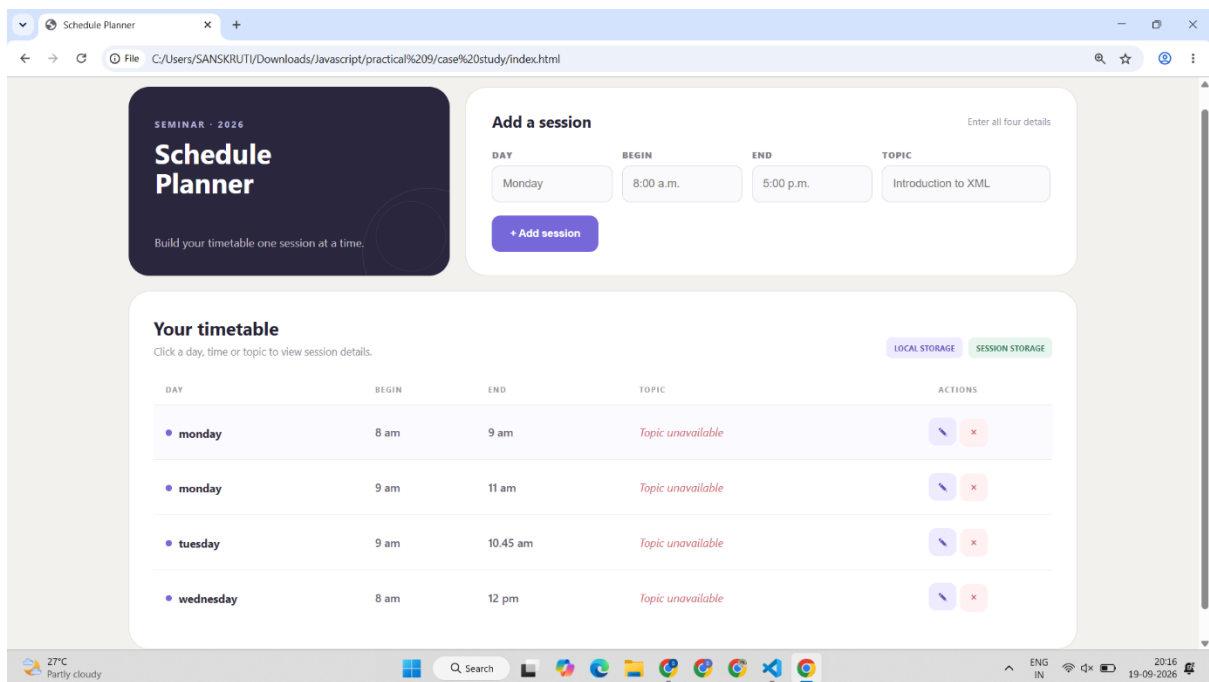
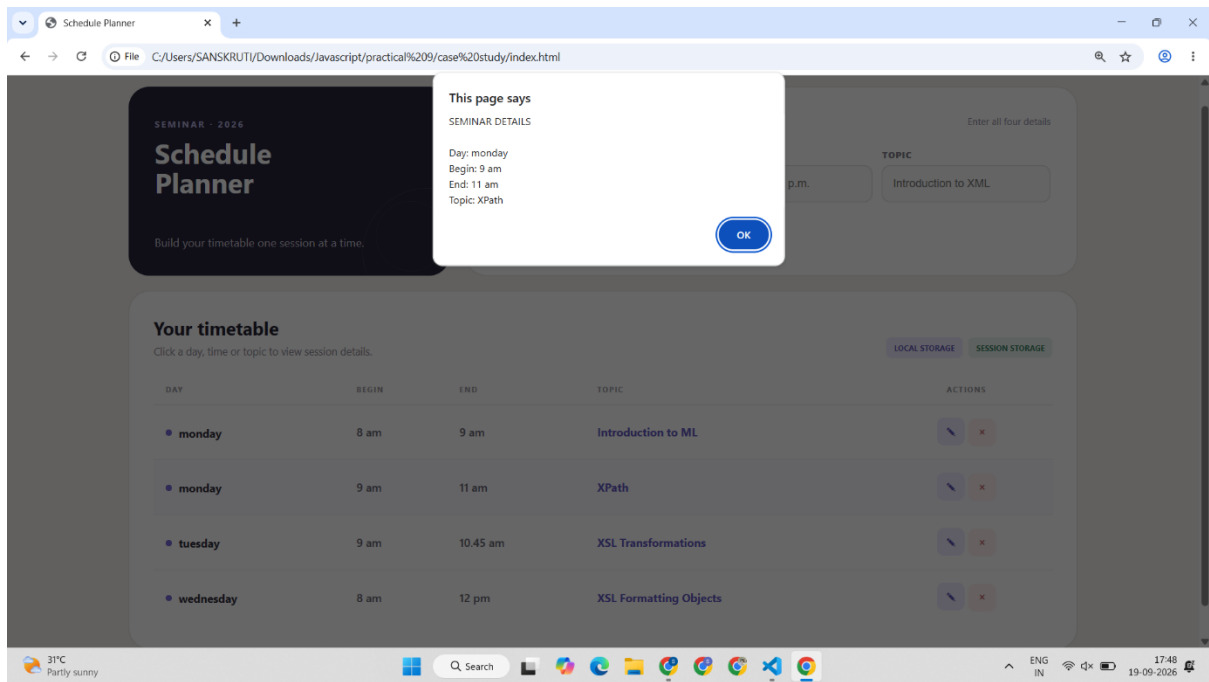
The screenshot shows a web browser displaying a 'Schedule Planner' application. The page has a dark blue header with the title 'SEMINAR - 2026 Schedule Planner' and a subtitle 'Build your timetable one session at a time.'.

On the right, there is an 'Add a session' form with fields for DAY (Monday), BEGIN (8:00 a.m.), END (5:00 p.m.), and TOPIC (Introduction to XML). A '+ Add session' button is below the form.

Below the form is a section titled 'Your timetable' with a subtext 'Click a day, time or topic to view session details.' and two tabs: 'LOCAL STORAGE' and 'SESSION STORAGE'.

DAY	BEGIN	END	TOPIC	ACTIONS
monday	8 am	9 am	Introduction to ML	
monday	9 am	11 am	XPath	
tuesday	9 am	10.45 am	XSL Transformations	
wednesday	8 am	12 pm	XSL Formatting Objects	

The bottom of the screen shows a Windows taskbar with the date 19-09-2026, time 17:48, and weather 31°C Partly sunny.



Conclusion –

The experiment successfully demonstrated **web storage** and **JavaScript click events** through theme preferences and a Schedule Planner.